DATA STRUCTURES AND ALGORITHMS

Lecture Notes 2

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ROAD MAP

- **Review Functions**
- Review Classes

Function

- A function is a group of statements that has a defined name that is used to refer to that group of statements so that they can be called.
- The groups of statements that make up a function perform a specific task.

Function

- Syntax

```
def functionname(parameter1, parameter2,… ):
    statements
```

- Example

```
def bmiCal(name , weight , height):
  bmi = weight / height ** 2
  print('Hello', name)
  print('Your BMI is' , bmi)
```

Function

```
def Function(n):
    i = s = 1
    while s < n:
        i = i+1
        s = s+i
        print("*")

Function(20)
```

- 1. What’s the result of this program?
- 2. How many times the command “print” running?

Classes

- Creating a class is an object-oriented language's ability to manipulate data to solve problems.
- There are two things involved:
  - refer to data or store what data (data members) and
  - what operations can be performed (member functions or methods).

Classes

- In python language, a new class is created by giving a name and defining a method associated with the definition.
- Syntax

```
class classname:
    methods
```

- After create class, we create instant or object variable to use class.

Classes

```
class Point:
   def __init__(self,x_input ,y_input):
     self.x = x_input
     self.y = y_input

myPoint = Point(3,7)
```

First method is constructor function.
In Python, constructor is called with __init__ (with two underscores).

Classes

- The three parameters are (self, x_input, y_input).
- self is a special parameter that will always be used as a reference to the object variable. It must always be the first parameter. However, it does not need to be set when called.

Classes

- self.x and self.y in the constructor are

data members of object variable that hold an internal data named x and y.

- Parameter values are initialized to x and y when the object variable is created.

Classes

- To create an instance of a Point class (or create an object variable), we do by taking the name of the class and passing it the necessary values (note that we do not call __init__ directly).
- For example:

```
myPoint = Point(3,7)
```

Classes

- The variable myPoint is an object variable of Class Point (it is a instant of Class Point).

```
class Point:
   def __init__(self,x_input ,y_input):
     self.x = x_input
     self.y = y_input
   def show(self):
     print(self.x, ' , ' ,self.y)

myPoint = Point(3,7)
myPoint.show()
```